

FYS5419/9419 February 4



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$$\hat{\rho}_A = \text{Tr}_B [\rho_{AB}]$$
$$= \langle 0_B | \rho_{AB} | 0_B \rangle + \langle 1_B | \rho_{AB} | 1_B \rangle$$

Discussion

$$p(x, y)$$

marginal probability

$$p(x) = \int_{y \in D} dy p(x, y)$$

$$|0\rangle_A = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$|1\rangle_A = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\langle 0_A | \rho_{AB} | 0_A \rangle$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\langle 1_A | \rho_{AB} | 1_A \rangle \Rightarrow$$

$$\hat{\rho}_A = (\hat{\rho}_B) = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2}$$

$$\text{Tr}[\rho_A^2] = \frac{1}{2} < 1$$

$$|\phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad \text{pure state}$$

$$\text{Tr} \rho^2 = 1 \quad (\text{Tr} \rho_{\phi^+}^2 = 1)$$

pure state

$$\text{Tr} \rho^2 < 1 \quad \text{mixed state}$$

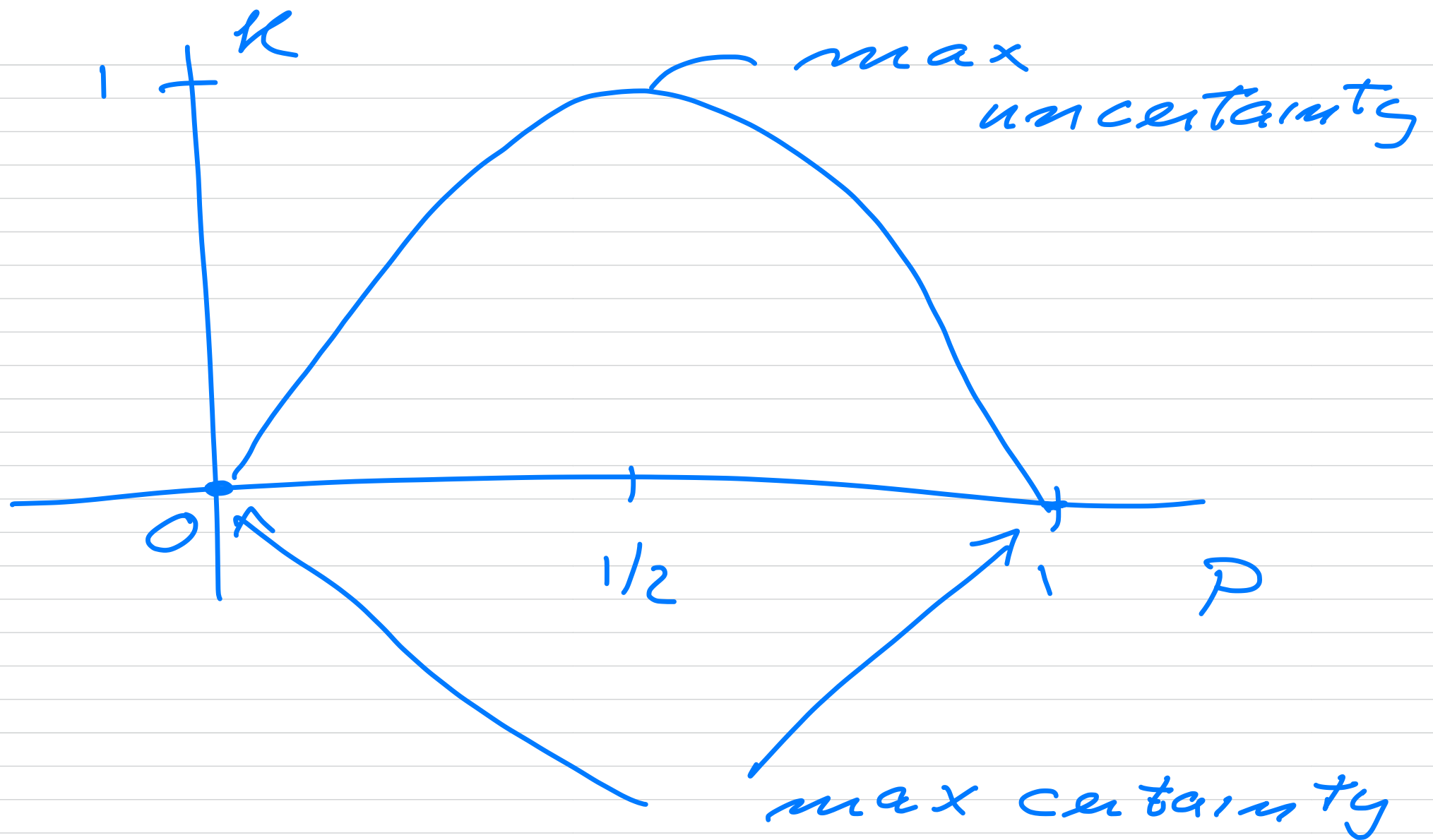
$$\mathcal{H} = - \sum_{i=1}^2 P_i \log P_i$$

$$\lim_{P_i \rightarrow 0} P_i \log P_i = 0$$

$$\mathcal{H} = - p \log p - (1-p) \log (1-p)$$

$$\mathcal{H} = 0 \quad \text{if } p = 0$$

$$\mathcal{H} = 0 \quad \text{if } p = \underline{1}$$



$$P(x,y) = P_x(x) P_y(y) \quad \log(P(x,y)) =$$

$$\log P_X(x) + \log P_Y(y)$$

$$\bar{I} = -\log p$$

← degree of uncertainty

